

SIWES REPORT GUIDE FOR THE DEPARTMENT OF MATHEMATICS

FACULTY OF NATURAL SCIENCES UNIVERSITY OF JOS

INTRODUCTION

• Modes of Examination:

The student will be examined on-site and internally

• Submissions:

For Internal Examination: The student is expected to submit a soft-bound copy of the report duly signed by his/her SIWES Coordinator a week to the date of the oral/internal examination.

• TYPING FORMAT

Students are advised to adopt the following typing formats.

Font Type and Size: use Times New Roman (for Microsoft Word) or the default Computer Modern/CM Roman (for LaTeX) as the font type and 12 as font size.

Paper Size and Margins: use A4 paper and leave 2.5 cm margin on the top, 1.2 cm at the bottom, 1.3 cm on the right and 3.8 cm on the left-hand side of the paper.

Line Spacing:

Use single line spacing for title page, approval page, dedication and within reference entries.

Use double line spacing for the acknowledgements, abstract, main body of the project and between references.

Pagination: the page number should appear at the center of the top of each page. Preliminary pages should be numbered in Roman numerals (i, ii, iii, …). For the rest of the work, starting from Chapter One, Arabic numerals (1, 2, 3, …) should be used.

NB: The student is at liberty to typeset his/her project using Microsoft Word or LaTeX.

COMPONENTS OF A COMPLETE SIWES REPORT

A complete SIWES report consists of the following components:

1. The Report Cover (Soft Bound)

2. Preliminary Pages

3. Main Work

4. References

5. Appendices (where applicable)

1. THE REPORT COVER

The front cover page must contain the following details:

• Title of the research

• Full name of the student (with surname last)

• Matriculation number of the student

• Course title

• Course Code

• Month and year in which the report is submitted for examination

2. PRELIMINARY PAGES

- Title Page
- Certification
- Dedication (Optional)
- Acknowledgements
- Abstract
- Table of Contents
- List of Tables
- List of Figures
- List of Abbreviations

NB: Students are expected to start each section on a new page.

Title Page

This page consists of the title of the SIWES Report, the name of the student, the matriculation number, the reason for the report and the month and year. See the sample below.

TITLE OF SIWES REPORT (IN CAPITAL LETTERS)

NAME OF STUDENT (IN CAPITAL LETTERS, WITH SURNAME LAST)

MATRICULATION NUMBER

LEVEL

COURSE TITLE

COURSE CODE

A SIWES REPORT SUBMITTED TO THE DEPARTMENT OF MATHEMATICS, FACULTY OF NATURAL SCIENCES, UNIVERSITY OF JOS, NIGERIA

MONTH, YEAR

Acknowledgements

Here, the student appreciates all that contributed technically, financially, morally and otherwise to the success of the report.

Dedication

The dedication page is where the researcher dedicates the report to a deity, someone, dead or/and alive.

Table of Contents

The titles of chapters, sections, sub-sections and their principal sub-division should be listed in the "Table of Contents" and must be worded exactly as they appear in the body of the report. For instance, Section 1., Sub section 1.1 and sub-sub section 1.1.1 respectively. The table of contents should be edited at the final stage of the report writing, to correctly capture the reflections in the work.

List of Tables/Figures

The list is to aid the reader in locating tables/figures. It should contain the tag numbers which reflects the content and the page numbers. It should be well-numbered and unambiguous.

LIST OF TABLES

TABLE	**PAGE**
1 One-way Analysis of variance	15
2 Laplace transform of some functions	23

LIST OF FIGURES

FIGURE	**PAGE**
1 Schematic of SIR model	14
2 Bayesian Posterior Model	39

NB: In the main content, the figures/tables should be well-labelled as Figure 1, Figure 2,… and Table 1, Table 2,…

Abstract

This is the synopsis or summary of the report. The abstract should allow a reader who is unfamiliar with the work to gain a swift and accurate impression of what the report is about. It is often written with the tense in past. The abstract should summarize the problem statement or motivation, specific objectives, the methodology employed, the findings, conclusion and recommendations in a coherent manner. It must be self-contained, self-explanatory and precise. This should be in a single paragraph without citations. The abstract should be approximately 100 to 200 words.

3. MAIN WORK

The main work consists of the following Chapters:

Chapter One: Introduction

- Background of SIWES
- Objectives of SIWES
- Objectives of Placement
- Scope of Work
- Duration
- Methodology

Chapter Two: Profile of Organization

- History
- Structure
- Vision and Mission

- Products/Services
- Departments
- Posting Unit
- Mathematical/Statistical activities

Chapter Three: Activities Performed

- Weekly activities
- Mathematical/Statistical tasks
- Data collection
- Data management
- Software used (Matlab, Maple, SPSS, R, Python, Stata, Excel, Power BI, SQL, etc.)

Chapter Four: Mathematical/Statistical Skills Acquired, Recommendations and Conclusion

- Data collection
- Data cleaning
- Results
- Visualization
- Interpretation
- Recommendations

CHAPTER ONE: INTRODUCTION

1.1 Background of SIWES

This section should provide a brief overview of the Students Industrial Work Experience Scheme (SIWES). The student should explain why SIWES was established, its importance in bridging the

gap between theoretical knowledge acquired in the classroom and practical experience in the workplace, and its relevance to the student's programme of study.

1.2 Objectives of SIWES

This section should state the general objectives of SIWES. It should explain how the programme helps students acquire practical skills, develop professional competence, understand workplace procedures, and apply academic knowledge to real-world problems.

1.3 Objectives of Placement

The student should state the specific objectives of undertaking the industrial attachment at the selected organization. These may include gaining practical experience in mathematical/statistical analysis, learning relevant software, understanding organizational operations, and developing professional and interpersonal skills.

1.4 Scope of Work

This section should describe the areas of work covered during the attachment. Students should indicate the major activities, duties, projects, analyses, software, and professional responsibilities they were exposed to, particularly those related to Mathematics, Statistics, Data Science, Computing, or their programme.

1.5 Duration

The student should state the starting and ending dates of the industrial attachment and, where appropriate, the total number of weeks or months spent at the organization. The student may also briefly indicate the working schedule or number of working days per week.

1.6 Methodology

This section should explain how the student carried out the activities and obtained practical experience. It may include observation, participation in assigned duties, interviews with staff, data collection, document review, practical exercises, statistical analysis, use of software, and supervision by workplace personnel.

CHAPTER TWO: PROFILE OF ORGANIZATION

2.1 History

This section should provide a brief history of the organization, including its year of establishment, founders or parent organization where relevant, major developments, and its present status and role.

2.2 Structure

The student should describe the organizational structure and reporting system. An organizational chart may be included to show the relationship between management, departments, units, and the student's posting unit. Diagrams can be inserted.

2.3 Vision and Mission

This section should state the organization's vision and mission statements and briefly explain how they guide its activities, operations, and services.

2.4 Products/Services

The student should describe the major products produced or services provided by the organization. The description should be related to the organization's main functions and activities.

2.5 Departments

This section should identify and briefly describe the major departments or units within the organization and explain their respective functions.

2.6 Posting Unit

The student should provide detailed information about the department or unit to which he or she was attached. This should include the functions of the unit, major responsibilities, staff roles, and the student's position or role within the unit.

2.7 Mathematical/Statistical Activities

This section should identify the ways Mathematics, Statistics, Data Science, or quantitative methods are applied within the organization. Examples include data collection, statistical analysis, forecasting, quality control, financial analysis, modelling, research, monitoring and evaluation, database management, reporting, and decision-making.

CHAPTER THREE: ACTIVITIES PERFORMED

3.1 Weekly Activities

The student should present a week-by-week account of activities performed during the attachment. Each entry should briefly state the tasks undertaken, skills learned, equipment/software used, and relevant outcomes.

3.2 Mathematical/Statistical Tasks

This section should describe specific mathematical or statistical tasks performed by the student. Examples include descriptive statistics, probability calculations, hypothesis testing, regression,

forecasting, quality control, optimization, sampling, database queries, statistical modelling, or data analysis.

3.3 Data Collection

The student should describe how data were obtained during the attachment. This may include questionnaires, interviews, observations, administrative records, databases, experiments, surveys, sensors, websites, or other organizational sources. The student should indicate the type of data collected and its purpose where appropriate.

3.4 Data Management

This section should explain how collected data were organized, entered, stored, cleaned, coded, checked, and prepared for analysis. Students should mention procedures such as handling missing values, identifying duplicates, correcting errors, coding variables, and maintaining data security.

3.5 Software Used

The student should identify the software and digital tools used during the attachment and briefly describe what each was used for. Examples include Excel for data entry, cleaning, calculations, tables, charts and basic analysis; SPSS for statistical analysis; R for statistical computing and visualization; Python for programming and data analysis; Stata for statistical/econometric analysis; MATLAB for mathematical computation and modelling; Maple for symbolic mathematics; Power BI for dashboards and reporting; and SQL for database querying and management. Students should focus on what they actually did with the software rather than merely listing software names.

CHAPTER FOUR: MATHEMATICAL/STATISTICAL SKILLS ACQUIRED, RECOMMENDATIONS AND CONCLUSION

4.1 Data Collection

The student should describe the practical skills acquired in planning, collecting, recording, and validating data. The student should explain how classroom knowledge of data collection was applied in the workplace.

4.2 Data Cleaning

This section should explain the student's experience in preparing raw data for analysis. This may include identifying missing observations, correcting errors, removing duplicates, checking inconsistencies, coding variables, and transforming data into an appropriate format.

4.3 Results

The student should summarize the major findings or outputs obtained from the analyses or activities performed. Results should be presented clearly using appropriate tables, statistical measures, models, or other relevant outputs.

4.4 Visualization

This section should describe the student's ability to communicate data using graphs, charts, dashboards, and other visual methods. Examples include bar charts, histograms, boxplots, scatterplots, line graphs, heatmaps, statistical plots, and Power BI dashboards.

4.5 Interpretation

The student should explain what the statistical results or mathematical outputs mean in practical terms. Emphasis should be placed on converting numerical results into meaningful information that can support organizational decisions.

4.6 Recommendations

The student should provide practical recommendations based on the experience gained during the attachment. Recommendations may address improvements in data management, statistical analysis, software utilization, workflow, training, equipment, supervision, or the SIWES programme itself.

4.7 Conclusion

The conclusion should provide a concise summary of the overall industrial attachment experience. The student should highlight the major knowledge and skills acquired, the relationship between academic training and workplace practice, challenges encountered, and the contribution of the attachment to professional development.

Suggested Format for Weekly Activities

Week	Activities Performed	Skills Acquired
Week 1	Orientation and introduction to the organization	Workplace procedures and professional conduct
Week 2	Data entry and organization	Data management
Week 3	Statistical analysis/Mathematical Analysis	Mathematical/Statistical analysis skills

Recommended Emphasis

Students should demonstrate how academic knowledge was applied to real workplace problems.

The report should, where applicable, show the progression:

Data Collection → Data Entry → Data Cleaning → Data Management → Statistical/Mathematical Analysis → Visualization → Interpretation → Decision-Making → Recommendations

NB: Students are expected to start each Chapter on a new page.

REFERENCES

- APA 7th Edition

The reference section consists of the list of all the sources arranged alphabetically that have been cited in the report. This includes all the relevant journals, books and all sources of information consulted in the report, either online or print.

Plagiarism should be avoided at all costs, all quoted and exact words of different sources should be properly referenced, in-text and at the references' list/bibliography.

References should be in single line spacing. APA (American Psychological Society), MLA (Modern Language Association), and Chicago or Turabian styles are the commonest referencing styles but the Department of Mathematics, University of Jos uses the APA style. The reference and the in-text citation formats are discussed below.

Reference Formats

The following are the reference formats expected to be adopted by a student when writing the list of references of a SIWES report.

• Journal/Article (with single author)

Surname, Initials of First name. Initial of Middle name (if available). (Year of Publication). Title of Journal/Article. *Name of Journal/Article (in italics), Volume Number* (Issue Number), Page range.

Example: John, P.A. (2026). Spatial Statistics of diabetes patients: A case study of Jos university teaching hospital. *Journal of Statistics, 20*(19), 1-18.

• Journal/Article (with more than one author)

Surname, Initials of First name. Initial of Middle name (if available)., Surname, Initials of First name. Initial of Middle name (if available). & Surname, Initials of First name. Initial of Middle name (if available). (Year of Publication). Title of Journal/Article. *Name of Journal/Article (in italics), Volume Number* (Issue Number), Page range.

Examples: Kallolkar, V.S. & Salunke, S.E. (2015). Laplace and Z-transformations. International *Journal of Scientific and Engineering Research, 4*(11), 12-18.

Textbook

Surname, Initials of First name. Initial of Middle name (if available). (Year of Publication). *Title of Textbook (in italics)*. (Editor/translator initials, Ed. or Trans.) (Edition if specified). Publisher, City, Country.

Examples: Lambert, J. D. (1973). *Computational Methods in Ordinary Differential Equations* (2nd ed.). John Willey and Sons, New York, USA.

Appendices

- Logbook (Snippet)
- Attendance (Snippet)
- Acceptance letter (Snippet)
- Weekly reports (Snippet)
- Outputs (Snippet)
- Questionnaires (If applicable) (Snippet)